ANMOL SRIVASTAVA

Backend Engineer · Python · FastAPI · RAG · LLM Integration

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SUMMARY

Backend engineer building production Python services with **FastAPI**, **PostgreSQL** and **Redis**, currently a full-time **Backend Engineer at BNCW** on Briefs, an AI stock-research SaaS platform for a US-based client. Built a GPU-accelerated OCR pipeline, an LLM routing gateway with failover and caching, RAG-based document Q&A, and an independently architected **5-service microservices backend** with JWT auth, rate limiting and API versioning. Open to backend, full-stack and AI / ML engineering roles.

EXPERIENCE

Backend Engineer (Full-time)

BNCW

09/2026 – Present | Dwarka Mor, New Delhi (On-site)

- Backend engineer on **Briefs**, a production AI stock-research SaaS platform for a US-based client: a monorepo of three **FastAPI** services (core API, RAG chatbot on **PostgreSQL + pgvector**, scoring engine) and a Next.js frontend, deployed to **AWS ECS** through CI-gated Bitbucket Pipelines with pre-prod → prod release promotion and Sentry monitoring
- Own the backend migration of newsletter ingestion from **Optipub to Customer.io** in the RAG chatbot service: an async HTTP API client with guarded cursor pagination and typed errors, an adapter that maps newsletters into the shared content model, and a read-only parity check that verifies both providers match before cutover

Backend Engineer Intern

Scalean Technologies

08/2026 | Pune, Maharashtra (Remote)

- Built a document OCR pipeline using PaddleOCR (GPU) achieving **1–2s response time** via GPU-accelerated inference and in-memory model loading; benchmarked 3 engines (Tesseract, EasyOCR, PaddleOCR) before selecting the optimal one
- FastAPI backend with async file handling and input validation; regex + bbox-based line grouping for structured field extraction (name, DOB, address)
- TypeScript frontend with live image preview, confidence score visualization, and real-time extraction results
- Independently architected and built a backend microservices system (**API Gateway + 4 services**) using FastAPI, PostgreSQL and React/TypeScript, implementing JWT-based authentication, rate limiting (SlowAPI), CORS configuration, and API versioning

Web Developer Intern

XeTech

06/2025 – 07/2025 | Noida, Uttar Pradesh (On-site)

- Built and styled responsive React.js UI components (JSX, props, state) for desktop and mobile within an existing project codebase
- Used Git and GitHub for version control: branching, committing, and pushing changes for team review

PROJECTS

SmartRoute – Intelligent LLM Gateway

github.com/AnmolSrivastavaaaa/SmartRoute

FastAPI · PostgreSQL · Redis · React · TypeScript · Groq · Gemini

- Architected a full-stack **LLM routing gateway** that classifies request complexity and routes across **2 providers and 3 cost tiers**, **cutting inference spend 58%**, with quality verified by an independent cross-provider benchmark rather than self-reported
- Built **cross-provider failover with a circuit breaker** (3 failures → 60s cooldown), validated by a live chaos test that revoked a real API credential mid-run and confirmed automatic recovery
- Implemented **tenant-isolated Redis caching**, cutting repeat-query latency **97% (1555ms → 42ms)**, plus per-key rate limiting via a Redis fixed-window counter
- Wrote **18 automated tests** covering routing, quality validation, and failover; built a **JWT-authenticated React + TypeScript dashboard** surfacing cost savings, cache hit rate, and p50/p95/p99 latency

SmartDocs – AI-Powered Document Q&A

github.com/AnmolSrivastavaaaa/SmartDocs-documents-with-AI-intelligence-LLM-python

- Built a private knowledge Q&A system: users upload documents, ask natural-language questions, and receive AI-generated answers grounded in their own files
- **Eliminated external LLM API cost and data exposure entirely** by running GPT-2 locally via Hugging Face and PyTorch, so documents never leave the user's machine
- Engineered the full RAG pipeline: document chunking, Sentence-Transformers embeddings, ChromaDB vector store, and semantic retrieval feeding LLM prompt construction

SKILLS

Languages

Python | JavaScript | TypeScript | Java | R

Backend

FastAPI | REST API Design | Async Python | Uvicorn | Microservices | JWT Auth | API Versioning | Rate Limiting | SQLAlchemy | Jinja2

Databases

PostgreSQL | Redis | ChromaDB | Vector Similarity Search

AI / ML

LLM Integration | RAG | Embedding Pipelines | Semantic Search | Sentence-Transformers | Hugging Face | PyTorch | GPT-2 / DistilGPT2 | Groq | Gemini API

OCR

PaddleOCR | Pytesseract

Frontend

React.js | HTML | CSS | Tkinter

Data & Libraries

Pandas | NumPy | Requests | SpeechRecognition | pyttsx3

Deployment & DevOps

Git | GitHub | Render | Railway | Cloud Deployment | Environment Configuration

Concepts

System Design | Automated Testing | Multithreading | Automation Scripting | Web Automation | Problem-Solving

EDUCATION

Bachelor of Computer Applications (IBM), Specialization in AI

United University

2023 – 2026 (Completed)

Prayagraj, Uttar Pradesh

LEADERSHIP

Founder & Band Leader, Music Band

United University

- Founded and led a college band from the ground up, handling member recruitment, rehearsal planning, and creative direction